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Test Procedure for Proportional Detectors

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1. OBJECTIVE AND SCOPE.

1.1 Objective.

The purpose of this document is to describe all the steps involved in the test procedure for proportional detectors.

1.2 Scope.

This procedure covers the test of small and large area flow gas proportional detectors (2π-α - β detectors).

2. NOTATION AND DEFINITIONS.

2.1. Notation.

MCA Multi Channel Analyzer.

2.2. Definitions.

2.2.1. Proportional detector.
A proportional detector or counter is a kind of gas filled detector which is capable of differentiating between various kinds of charged particles: α and β radiation.

2.2.2. Background.
Events registered by the detector per unit time from radioactive decay occurring in the environment or from cosmic rays interactions.

2.2.3. Activity.
The number of nuclear decays (transformations) per unit time. The measurement unit is the Becquerel (Bq), where one Bq is equal one decay per second (s^{-1}). The Curie (Ci) is also used and equals to 3.7×10^{10} Bq.

2.2.4. Alpha emission.
The number of alpha particles emitted per unit time from a source with alpha-emitting radionuclide.

2.2.5. Beta emission.
The number of beta particles emitted per unit time from a source with beta-emitting radionuclide.

2.2.6. P-10 gas.
A gas mixture commonly used in alpha/beta proportional counter detectors, consisting approximately of 90 % argon and 10 % methane.

2.2.7. Radioactive source.
A sample that contain one or more radio nuclides whose activity, concentration, emission rate or surface emission rate of ionizing radiation can be measured.

3. DEVELOPMENT.

The content of this document describes all the steps required to test a proportional detector. A flow chart is shown in the Annex I which summarizes this process.

3.1. Recommended equipment and performance requirements.

The list of all recommended equipment required for testing of proportional detectors is shown below. Alternative equipment may be used as long as the substitute equipment has specifications at least as good as those listed. Additionally ensure that all incoming support equipment has a current calibration or calibration check.

3.1.1 Charge sensitive Preamplifier
3.1.2 P-10 counting gas cylinder, regulator and control valve.
3.1.3 Amplifier.
3.1.4 A scaler, a multi-channel analyzer (MCA) or a counting system.
3.1.5 A DC high voltage power supply.
3.1.6 Oscilloscope.
3.1.7 A source with alpha or beta emitting radio nuclide (Am-241, Sr-90, Po-210, etc). A source with an activity of around 37000 Bq (1 μCi) could be used.

3.2. Preliminary operations.

Before beginning the test of the proportional detector, make sure that the proportional detector is working correctly.

3.2.1. Inspection for damage (mainly the Mylar window).
If damage such as broken Mylar window or seal, dented packages, etc. has occurred, the proportional detector will not be tested and a report of the fault conditions will be written.

3.2.2. Record on the Annex 3 the information about the proportional detector:
- Manufacturer.
- Model and serial number.
- Date of reception.
- Operating voltage, etc.

3.2.3 Filling gas.
The filling gas in a proportional counter is usually a noble gas because noble gases are not electronegative and do not react chemically with the detector components. Argon is the gas most widely used because its low cost. Other noble gases with higher atomic numbers (for example: krypton and xenon) might be used if an increase in the sensitivity to X rays or gamma rays is required.

Hydrocarbon gases (for example: methane, propane and ethylene) can also serve as a filling gas, but they have the disadvantage of being flammable. P-10 gas, developed by John Simpson in the 1940s, is the most widely employed gas for gas flow proportional counters.

3.2.4 Counting gas connections.

i) Mount the regulator on the cylinder and turn the pressure adjustment knob counter clockwise for minimum pressure.

ii) Connect the proportional detector to the fill gas tank using clear plastic tubing, as shown in the Figure A of Annex II.

iii) Open the high pressure bottle valve on the cylinder and set outlet pressure to 5 psi using the regulator valve.

iv) Adjust the flow control valve to approximately 20 cm^3/min. This flow can be changed to match the requirements of different kinds of proportional detectors.

v) The gas flow system must be operating at least for 30 minutes before applying the high voltage to the detector.

3.2.5 Preliminary test for proportional detectors.

i) Connect the proportional detector, amplifier, scaler, high voltage power supply and oscilloscope, as shown in the Figure B of Annex II.

ii) The amplifier gains and threshold voltages are set in the middle of the interval.

iii) Turn on the high voltage power supply and other electronics.

iv) Place a radioactive source on the proportional detector. A beta source may be used to test the detector after it has been flushed completely with the P10 counting gas.

v) Increase the high voltage to get signal pulses on the oscilloscope.

vi) Measure the signal pulse size. It depends of three parameters:

Operating voltage.

When a detector is biased with a very high voltage, a large avalanche is formed inside it and as a consequence, big pulses are produced.

Energy deposited into a proportional detector.

When incident particles deposit high energy in the detector gas, it generates a large number of primary ion pairs, producing a large avalanche, consequently big pulses are obtained.

Kind of particles.

In proportional detectors, pulses produced by alpha particles are larger than the pulses produced by beta particles or gamma rays.

v) The Fig. 1 shows a typical signal pulse from a proportional detector.

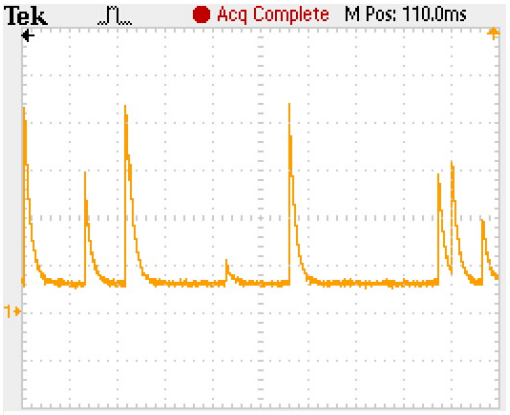


Fig. 1. Proportional detector pulses as seen on the oscilloscope screen.

3.3. Test of the proportional detector.

Testing a proportional detector involves the determination of the operating voltage and measurement of the background.

3.3.1 Operating voltage for a proportional detector.

In order to determine the proper operating voltage for a proportional detector, the counts vs. applied voltage graph (the plateau graph) needs to be obtained The length of the plateau curve depends of the kind of detector and the counting geometry. Therefore, it can be obtained with the following steps:

i) Verify that the P-10 gas flows through the system.
ii) Place a radioactive source on the center of the proportional detector, in order to generate signal pulses. The geometry must be the same for all the measurements.
iii) Increase the high voltage by steps and read or calculate the count rate.
iv) Record these values in the Table 1 of Annex III.
v) Plot the counts vs. voltage in the graph.
vi) Define the operating voltage (V_o) in the middle of the plateau, according to the equation:

$$V_O = V_1 + \frac{V_1 - V_2}{2} \qquad (1)$$

Where:
V_1 is the initial voltage in the plateau
V_2 is the final voltage in the plateau

vii) See the Fig. 2 and Fig. 3 for typical plateaus.

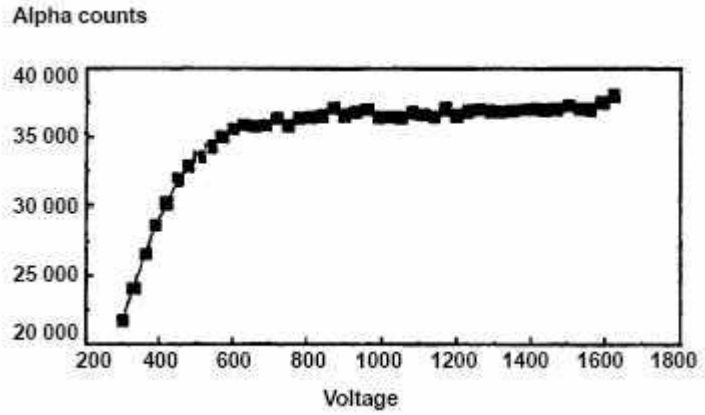


Fig. 2. An alpha plateau measured with a Po-210 source.

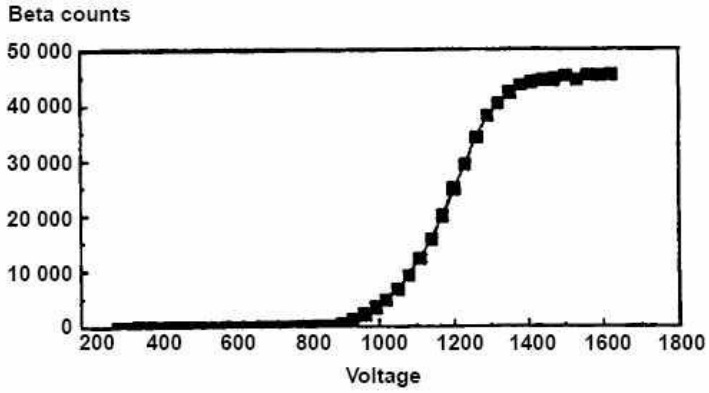


Fig. 3. A beta plateau measured with a Sr-90 source.

3.3.2 Background.

There are two kinds of background measurements in the proportional detectors: long term (quarterly) and short term (daily) measurements.

i) Long term measurements.

These measurements should be carried out for a long period of time (several hours) and help to avoid biases caused by the diurnal variation of the background. The background obtained from these measurements shall be used if background subtraction is required. See Fig. 4 for an example of the diurnal variations.

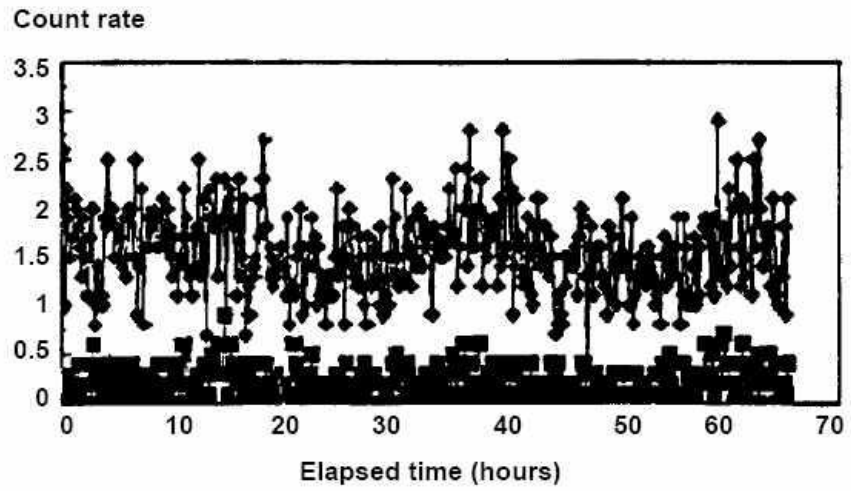


Fig. 4. Alpha and beta background count rates (CPM) over a long period of time.

ii) Short time measurements.

These measurements are used as a quality control check to verify that the system has not become contaminated and it is performed within acceptable limits. The measurements should be acquired for a period of time comparable with the counting time of the samples.

In this test procedure, the background is measured for one hour daily. Record these values in the Table 3 of Annex III and plot the counts vs. time graph. The information should be filed as a quality control for the proportional detector.

3.4 Troubleshooting.

3.4.1 Loss of gas through a ruptured Mylar window. If the rupture is small, a piece of transparent tape may be applied to the hole.

3.4.2 Excessive count rate may be caused for:

i) Radioactive contamination of the detector. Contamination may usually be removed with a soft brush or water rinse.

Caution: The water should not be allowed to enter the interior of the detector.

ii) Dirt or moisture in the coaxial connector or isolator materials. Clean this connector with methanol and dry with hot air.

iii) Moisture inside the detector can cause corrosion on the anode wire supports, and it can produce spurious electrical noise. To clean inside the detector a Freon type solvent may be used and the Mylar window must be replaced if it is needed.

4 ADMINISTRATION OF THE REPORTS

4.1 Test report.

The test report must be elaborated by the operative personnel. This document summarizes all the tests carried out with the proportional detector.

4.1.1 The test report should include the following components: In the first part a descriptive title, identification number, information about the detector and test equipments. The second part is the body of the technical report that includes: measurement results, graphics, conclusions and discussion. Finally the third part has the issue date, author name and reviewer name. The Annex III shows an example.

4.1.2 The test report must be reviewed by the area responsible.

5.- ACTION IN CASE OF NON CONFORMITIES.

5.1 Technical Report.

Even in the case that results of the test are not as expected, a technical report has to be elaborated, indicating the non conformities and how far are the measured characteristics from the ideal ones.

5.2 Labelling.

The components or equipments that are not under specifications or with a failure have to be marked with a label indicating: **OUT OF SPECIFICATIONS** and **FAILURE** respectively.

6. RESPONSABILITIES.

6.1. Head of the Department.

The application of this procedure should be supervised by the head of the department.

6.2. Area responsible.

6.2.1. Assure that the test equipments are in good operational conditions and additionally assure that all equipments are calibrated.

6.2.2. Supervise the activities for testing of proportional detectors.

6.2.3. Verify the test report.

6.3 Operative personnel.

6.3.1. Verify that the test equipments are in good operational conditions and additionally verify that all equipments are calibrated.

6.3.2. Apply the test procedure for proportional detectors.

6.3.3 Elaborate the test report.

7. BIBLIOGRAPHY.

1 ANSI N42.25-1997 "American National Standard Calibration and Usage of alpha/beta proportional counters", January 6, 1997, USA.

2 Knoll, Glenn F. "RADIATION DETECTION AND MEASUREMENT", Third Edition, John Wiley and Sons. U.S.A. 2000

3 IAEA, "Calibration of Radiation Protection Monitoring Instruments", Safety Report Series No. 16, Vienna 2000.

4 ASTM "Standard Practice for Set-up, Calibration, and Quality Control of Instruments Used for Radioactivity Measurements" ASTM D7282-06 Standard, 2006.

8. ANNEXES.

Annex I.
Flow chart.

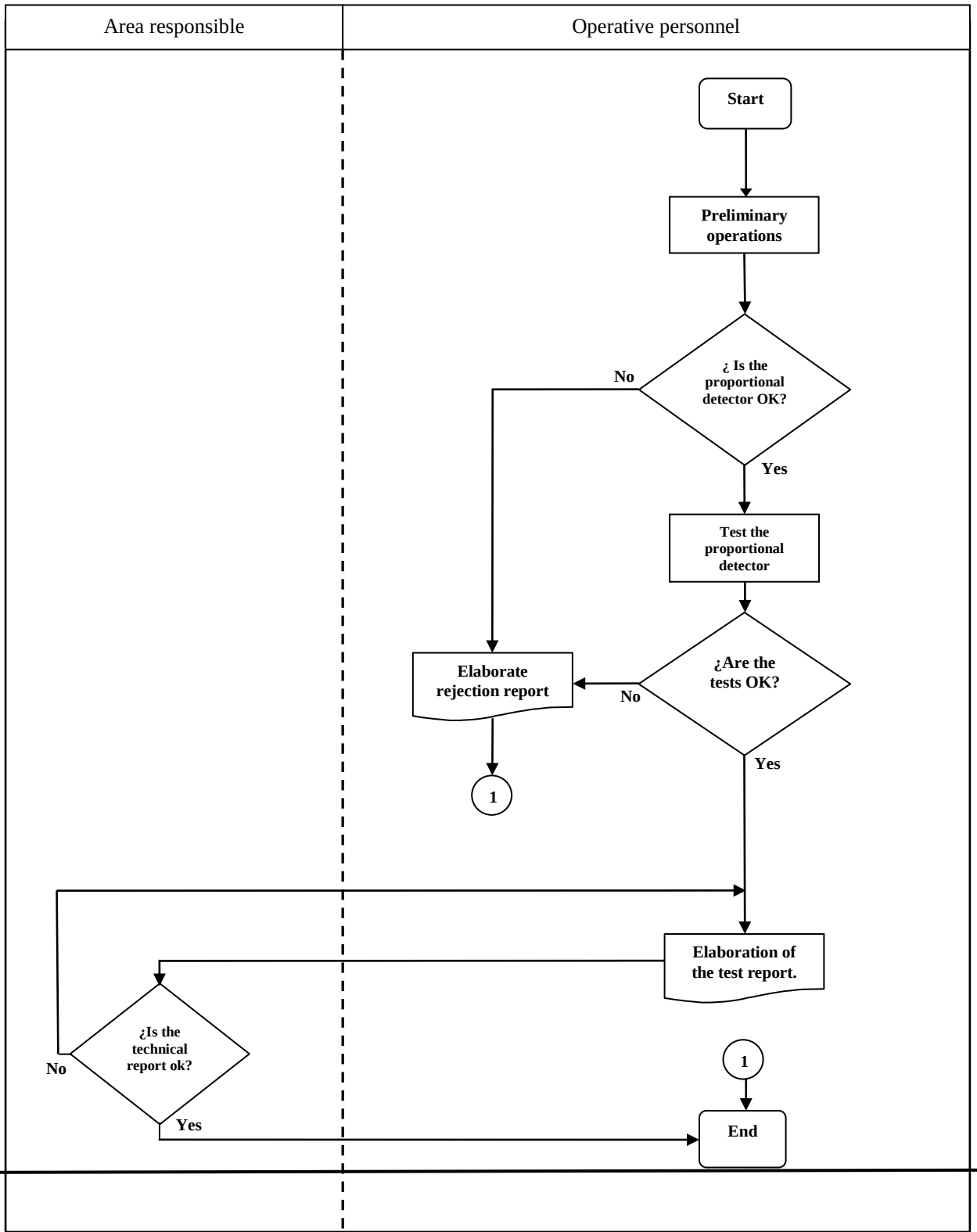
Annex II.
Fig. A. Set-up of the gas connections.

Fig. B. Connections for the test of a proportional detector.

Annex III.
Test report.

Annex I.

Flow chart.



Annex II.

Fig. A

Fig. B

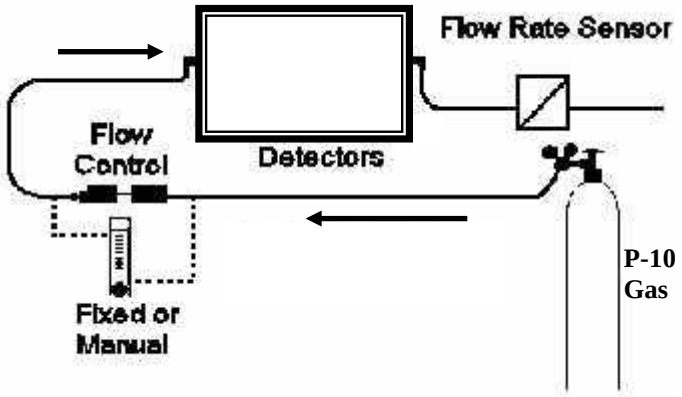


Fig. A. Set-up of the gas connections.

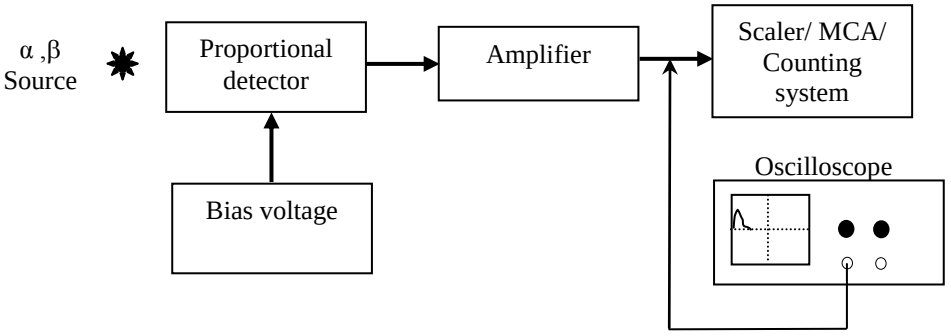


Fig. B. Connections for the test of a proportional detector.

Annex III

Test Report

TEST REPORT No. PD - _____

Proportional detector

Manufacturer: _____ Model: _____

Serial number: _____ Window: _____

Test equipments

Manufacturer: _____ Model: _____

Serial number: _____

Manufacturer: _____ Model: _____

Serial number: _____

Radioactive source

Source: _____ Energy: _____

Activity: _____ Date: _____

Results.

1. Operating voltage for a proportional detector.

Table 1. Readings to obtain the plateau graph.

Item	Voltage (V)	CPM
1		
2		
3		
4		
5		
6		
n		

n = number of readings are proposed by user

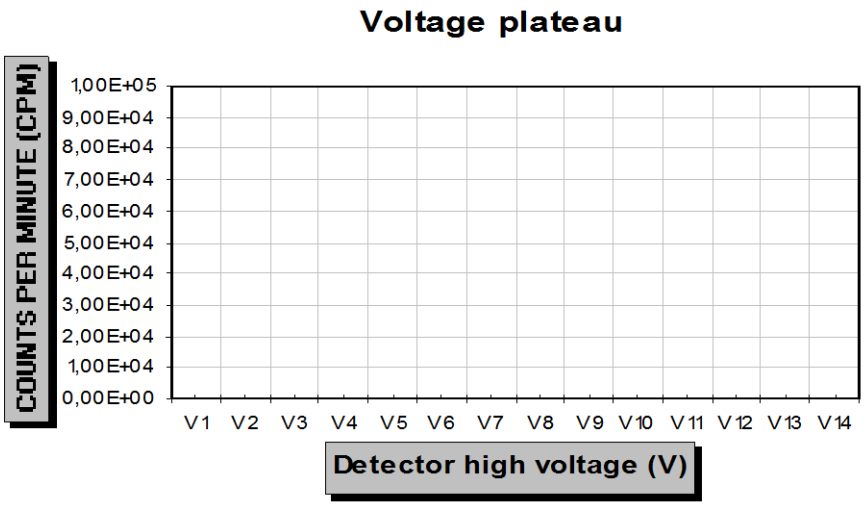


Table 2. Operating voltage.

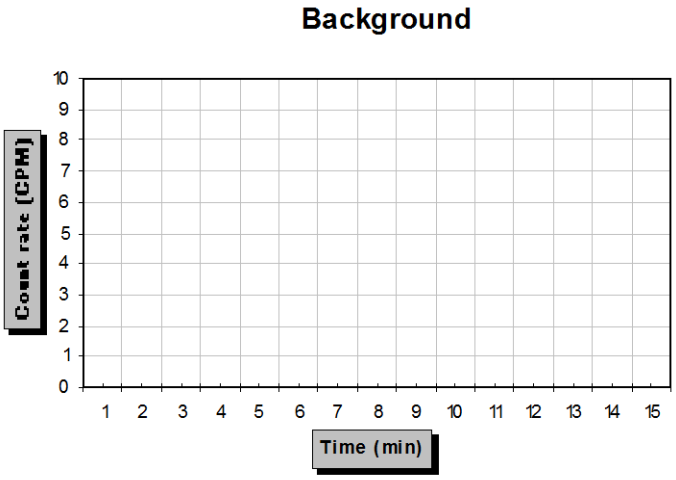
Plateau voltage	
Item	Measured value
V_1 =	
V_2 =	
Operating voltage	
	Calculated value
V_O =	

2. Short term Background.

Table 3. Background readings.

Elapsed time (min)	Count rate (CPM)
1	
2	
3	
4	
5	
.	
n	

n = One hour = 60 min.



3. Diagnostic or conclusions.

Tested by: _____

Reviewed by: _____

Date: _____